

Rehan Vipin

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Work Experience

Morgan Stanley

Aug 2022 - Present

Software Engineer 3

- Leading a team automating data pipelines for financial models, saving controllers 100+ hours per quarter. Architected the system by collaborating with business, product, and tech stakeholders.
- Designing a python modelling framework that enables analysts to develop and productionize models 5x faster.
- Delivered a self-service data transformation app used by 60+ modellers, reducing data preparation time by 10x.
- Led platform modernization in the dept., including a move to cloud-ready deployments, saving 30k+ \$ in a year.
- Built internal dev tools and AI workflows that are used daily across the dept., saving 200+ hours / quarter.
- Identified and optimized a computation bottleneck in the forecasting engine. Profiled and revamped the multi-threaded Java process to achieve a 90% reduction in runtime.

Tech stack - Angular, Spring Boot, DB2, Python, Docker, Snowflake

Morgan Stanley

Jan - July 2022

Technology Intern

- Built a visualizer for financial analysts to understand data flows between 100s of models, 5x faster than before. Designed the UX by collaborating with business for reviews and iterations.
- Created a generic TS library for devs to make their own dependency visualization UIs in less than 20 mins.

Tech stack - Angular, d3js, TypeScript

Morgan Stanley

May - July 2021

Technology Intern

- Created a web app to analyze large datasets (1M+ rows and 20+ cols) in the browser.
- Reduced initial load time by 75% and browser memory usage from O(n) to O(1).

Tech stack - AG Grid, Spark, Spring Boot

PES Innovation Lab

May - July 2020

Research Intern

- Designed a framework for creating privacy-preserving neural networks using secure multi-party computation.
- Decreased inference time of models by 25% compared to the state-of-the-art garbled circuits framework.
- Added circuit approximations, cryptographic optimizations and automated model pruning as optimizations.

Tech stack - C, Python

Education

PES University, Bengaluru

July 2018 - Aug 2022

B.Tech. Computer Science Engineering.

Graduated with 9.11 CGPA, and an additional specialization in Systems and Core computing.

Technical Skills

Domains: Software Architecture, Distributed Systems, Parallel Computing, Cryptography, Deep Learning.

Front-End: TypeScript, JavaScript, Angular, D3js, Figma UX design.

Back-End: Python, Java, C, C++, CUDA, Flask, Spring Boot.

Infrastructure: SQL, Bash, DB2, MongoDB, Snowflake, Docker.

DevOps: Jenkins, GitHub Actions, Sourcegraph, Moderne, Copilot Cloud Agent, OpenTelemetry.

Projects

Personal projects and small-team projects.

GPU-optimized cryptographic hashing

May 2021

Rewrote the fastest crypto hash function, BLAKE3, with a new design to run efficiently on Nvidia GPUs. Optimized the parallel algorithms using CUDA C++ and SIMD.

Parallel and Distributed Heterogeneous Compute Platform

June 2021 - March 2022

Built a python framework to speed up compute-heavy tasks by rewriting them with building blocks that are processed and executed in a distributed + parallel fashion.

std::skip_list

Nov 2020

Implemented a C++ STL compatible Skip Lists library. Profiled operations on the data structure to visualize performance, and added user guide docs.

Faster CNNs using FFTs

May 2020

Developed a Python library for faster convolutional neural networks, based on the principle of Fast Fourier Transforms. It was 20% faster than the Tensorflow implementation of CNNs at the time (2020).

Additional Experiences

Research Intern, Machine Learning - Ennoventure

Jan - Feb 2021

- Researched and trained prototype models as part of a shift towards Hidden Markov Models and Kalman filters for better quality predictions.
- Made a stock market predictor to demonstrate feasibility of the solution.

Teaching Assistant, Unix - PES University

Jan - May 2020

- Incorporated hands-on challenges to teach OS concepts and scripting. First time in the course history.
- Prepared the syllabus and assignments, and evaluated the students' work.

Subject Matter Expert, Blockchain - PESU IO

Sept - Nov 2019

- Taught Decentralized applications and Blockchain fundamentals in a 6-week peer-to-peer program.
- At the end of the course, students made custom decentralized apps from scratch.
- Additionally, mentored students individually, to help overcome blockers.

Awards & Recognition

Winner of Gen AI Challenge, Morgan Stanley, 2025

Awarded for the best use of Gen AI in development in a competition with 100+ participants worldwide.

Rookie Talent, Morgan Stanley, 2023

Awarded for most impactful project contributions amongst the new developers in the Division.

Semi-Finalist, e-Yantra, 2019

Top 100 nationally amongst 7000+ teams. Built a disaster relief supply bot using an embedded-C microcontroller and image recognition.